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Developments in radar and remote-sensing methods for measuring and forecasting rainfall

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Over the last 25 years or so, weather-radar networks have become an integral part of operational meteorological observing systems. While measurements of rainfall made using radar systems have been used qualitatively by weather forecasters, and by some operational hydrologists, acceptance has been limited as a consequence of uncertainties in the quality of the data. Nevertheless, new algorithms for improving the accuracy of radar measurements of rainfall have been developed, including the potential to calibrate radars using the measurements of attenuation on microwave telecommunications links. Likewise, ways of assimilating these data into both meteorological and hydrological models are being developed.

In this paper we review the current accuracy of radar estimates of rainfall, pointing out those approaches to the improvement of accuracy which are likely to be most successful operationally. Comment is made on the usefulness of satellite data for estimating rainfall in a flood-forecasting context. Finally, problems in coping with the error characteristics of all these data using both simple schemes and more complex four-dimensional variational analysis are being addressed, and are discussed briefly in this paper.

Keywords: radar; remote sensing; data assimilation; rainfall; flood forecasting

1. Introduction

Over the last 25 years or so progress in the numerical prediction of rainfall has been slow compared with the predictions of other parameters. While operational hydrologists use numerical weather prediction (NWP) forecasts to provide warnings of the likelihood of flood conditions on time-scales beyond 6 h or so ahead, the basis of shorter-period forecasts depends upon the acquisition and use of observational data, particularly rain-gauge and radar data, although satellite data are of importance in data sparse areas. Increasingly, data from mesoscale numerical models are being blended with observational data, and remotely sensed data are being inputted to NWP, to improve both measurements and forecasts of precipitation (Golding 2000).

One contribution of 18 to a Discussion Meeting 'Flood risk in a changing climate'.

This interdependence of remotely sensed data and NWP models and their outputs is increasingly driving the further development of radar and other remote-sensing methodologies for measuring and forecasting rainfall. Palmer (2001) notes that conventional parametrization schemes in weather- and climate-prediction models describe the effects of sub-grid-scale physical processes. Indeed, he suggests that the remaining errors in these models may have their origin in the neglect of grid-scale variability, and he proposes that such variability should be parametrized by non-local dynamically based parametrization schemes.

A complementary approach to the parametrization of sub-grid-scale variability is the assimilation of remotely sensed observational data explicitly representing such variability using complex numerical schemes (see, for example, Marécal *et al*. 2001). The problem with such an approach is that these data contain errors which may cause numerical integrations to become unstable whatever numerical methods are used. Hence small differences in input data may lead to very different forecast outcomes. This problem is recognized, and is one reason for the major effort being devoted to the development of ensemble prediction schemes (e.g. Trevisan *et al*. 2001). It is also providing fresh impetus to efforts to improve the quality of radar estimates of precipitation, although this need is also driven by the requirements of nowcasting (forecasts from 0 to 3–6 h ahead).

It is now over 40 years since digital weather-radar data became available in significant quantities, and yet, while there has been considerable improvement in the quality of these data, flood forecasters remain reluctant to use them directly as input to river-catchment models. This is because, as for NWP models, errors in the data detrimentally impact flow forecasts (see, for example, Collier & Knowles 1986; Pessoa *et al*. 1993). It is worth noting in passing that the introduction of advanced computing technologies into hydrology, such that more complex distributed models can be run cost effectively (Apostolopoulos & Georgakakos 1997), will not in itself lead to improvements in operational flood forecasts due to this error-propagation problem.

In spite of the difficulties in using remotely sensed data, it is clear that major improvements in rainfall and flood forecasting are only likely to come through improvements in these data and the ways in which they are used. In what follows, we discuss improvements in radar technology and data processing linked to other remote-sensing systems and the degree to which they have improved, or promise to improve, measurements and forecasts of rainfall.

2. Developments in the use of single-frequency, single-polarization radars

(*a*) *Recent hardware developments*

Doppler radars, having the ability to measure the radial motion of target hydrometeors as well as the signal backscattered from them, are now standard. This follows the rapid growth worldwide in the operational deployment of digital radar technology over the last 20 years or so. More recently, digital receiver technology has been introduced.

A digital intermediate-frequency (IF) receiver accepts the analog IF signal (typically 30 MHz), processes it and outputs a stream of wide-dynamic-range digital in-phase and quadrature-phase signals which are then processed. In addition, the digital receiver can accept the transmit pulse 'burst sample' for the purpose of measuring the frequency, phase and power of the transmit pulse.

The digital approach replaces much of the traditional IF receiver components with flexible software-controlled modules and has the following advantages:

(i) a wide-dynamic-linear-range-improved receiver performance;

(ii) improved phase stability at long range for magnetron (as in Europe) systems;

(iii) adaptable algorithms;

(iv) reduced cost of spares;

(v) reduced scheduled maintenance costs; and

(vi) reduced mean time to repair (MTTR) and higher system availability.

Hence, as well as improving receiver performance, the use of the digital technology also enables provision of Doppler capability almost as a byproduct at small extra cost. In addition, it is now straightforward to monitor continuously, and set alarms as appropriate, transmitted power, receiver noise and the power of a stable noise source, so that a full calibration of the system may be performed automatically (e.g. Manz *et al*. 2000). Nevertheless, it remains difficult to accurately determine the radar-calibration constant (leading to radar-calibration bias), and novel methods using satellite-borne instrumentation have recently been described (Anagnostou *et al*. 2001).

(*b*) *Retrieving reliable precipitation data*

The spherical geometry of weather-radar scans results in the separation of data points in one part of the analysis domain, causing them to differ from those in another. Hence, schemes have been developed to transfer the data to uniform Cartesian grids to enable easier collocation with maps and digital terrain datasets. While this process is widely used—and, in the main, is not conceived to be causing any problems with the data—recent work by Trapp & Doswell (2000) points out that unambiguous assessment of physical evolution may be detrimentally affected by the process.

Much more serious problems in interpreting radar data (measured radar reflectivity of the targets, Z) in terms of rainfall are related to problems with the basic data themselves. The introduction of Doppler technology has enabled significant improvements to be made in the removal of ground clutter echoes, including anomalous propagation echoes, although there still remains a need for complementary techniques. Operationally, Doppler clutter cancellation is most effective when combined with techniques based upon either maps locating ground echoes and/or statistically based procedures (see, for example, Grecu & Krajewski 2000).

While the use of S-band (wavelength 10 cm) frequency, as in the USA, removes problems associated with the attenuation of the radar beam as it passes through precipitation, the use of C-band (wavelength 5 cm) frequency by much of the rest of the world to gain the advantage of improved sensitivity continues to be a source of measurement problems. Also C-band systems are presently a factor of two cheaper than S-band systems, although this situation may change with the introduction of

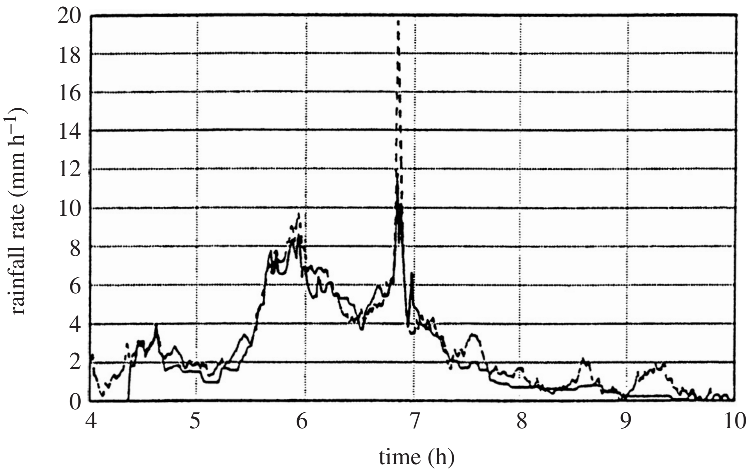

Figure 1. Time-series of rainfall estimates from rain gauges (——) and microwave link (– – –) attenuation measurements on 10 February 2000 (from Holt *et al*. 2000).

tunable travelling wave tube (TWT) technology. Correction procedures have been developed for attenuation at C-band, but problems remain (Capsoni *et al*. 2001).

For some years it has been recognized that errors in estimates of surface precipitation arise from the difficulty of converting radar reflectivity to precipitation when the radar beam intersects the bright band (the region where snow melts to form rain) or hail, and when evaporation or growth occurs below the radar beam (for a summary see Collier (1996)). This led to operational attempts to get around these problems using rain-gauge adjustment techniques (see, for example, Collier *et al*. 1983). However, it was only comparatively recently, building on the work reported by Joss & Waldvogel (1990), that the full significance of the need to derive and adjust the vertical profile of reflectivity (VPR) was recognized. A range of procedures to carry out adjustments, based on the radar data themselves and models of the physical processes involved in generating the profile, have been developed over the last 10 years or so (Smith 1986; Kitchen & Blackall 1992; Kitchen & Jackson 1993; Hardaker *et al*. 1995; Andrieu & Creutin 1995). Direct measurements of the VPR have also provided improved understanding of the problem (see, for example, Fabry *et al*. 1992). It is now generally accepted that the VPR must be adjusted before retrieving surface-precipitation estimates. The appropriateness of subsequent rain-gauge adjustment of the radar estimates to mitigate residual errors remains uncertain, although the application of time-integrated rain-gauge data does seem to produce improvements, particularly in mountainous terrain (Joss & Lee 1995). The use of statistical techniques to describe the spatial continuity of precipitation in such regions has also been shown to be advantageous. Germann & Joss (2001) use variograms of radar effectivity as a summary statistic to describe the spatial continuity of Alpine precipitation. This allows the representativeness of a point rain-gauge observation to be quantified.

A possible alternative or complement to rain gauges for adjusting radar data could be the use of path-averaged rainfall obtained from the difference in attenuation at two frequencies along a microwave path (Hardaker *et al*. 1997). Recent field trials have confirmed that good agreement can be achieved between the path-averaged rainfall and rain-gauge estimates of rainfall (figure 1), although the attenuation measurements are adversely affected by the occurrence of melting snow.

3. Developments in the use of multiparameter radar

(*a*) *Recent hardware developments*

Development of multiparameter radar hardware has been slow since the initial production of high-speed switches enabling the rapid alternate transmission of horizontally and vertically polarized microwave radiation described by Seliga & Bringi (1976). However, in recent years—following work on other polarization states such as circular (see, for example, McCormick & Hendry 1975) and the recognition of the potential of multiparameter radar for measuring precipitation—attention to the design of the hardware has increased.

Versatile research radars such as the CSU-CHILL installation (Brunkow *et al*. 2000) have provided the test beds from which to consider what polarization base is most effective in measuring rainfall and hydrometeor type. Scott *et al*. (2001) present observations of the standard dual-polarization meteorological quantities (Z_{DR}, ϕ_{DP} and ρ_{HV} defined later) measured from *simultaneous* horizontal (H) and vertical (V) transmissions. Because the different polarizations are simultaneous, no high-power polarization switch is needed. The relative phases of the H and V components were such that the transmitted polarization was circular. This circular polarization is shown to detect horizontally oriented particles such as rain with the same effectiveness as linearly polarized transmissions, and optimally detects randomly oriented or shaped particles such as hail. Circular polarization also optimally senses non-horizontally oriented particles such as electrically aligned ice crystals. They concluded that because the H and V transmissions are simultaneous it is feasible to make polarization-diversity measurements by transmitting other orthogonal polarizations on successive pulses, such as left-hand circular and +45° slant linear.

This form of simultaneous transmission is now being implemented on the NSSL's (National Severe Storms Laboratory) research WSR-88D S-band radar, and is being considered as the basis of the polarimetric upgrades to the operational WSR-88D radars in the USA (Doviak *et al*. 2000). However, it is not yet clear exactly which polarization states are optimum for precipitation measurements and work is continuing. Recently, Torlaschi & Gingras (2000), using a simple parabolic model of a convective rain cell (peak rain rate 50 mm h^{-1}), showed that the use of simultaneous transmission and reception of linear vertical and horizontal polarization at S-, C- and X- (wavelength 3 cm) bands provides accurate estimates of the intrinsic scattering properties of precipitation (figure 2). However, an analysis of the bias on the radar observables shows that all the observables, with the exception of reflectivity, can be severely affected.

(*b*) *Retrieving reliable precipitation data*

Much of the early work with polarization diversity radars was carried out at S-band. This choice of wavelength simplifies analysis as the along-path attenuation through rain is negligible as mentioned in the previous section. In addition to the radar reflectivity, usually horizontally polarized (vertically polarized in the UK), the following parameters may be estimated:

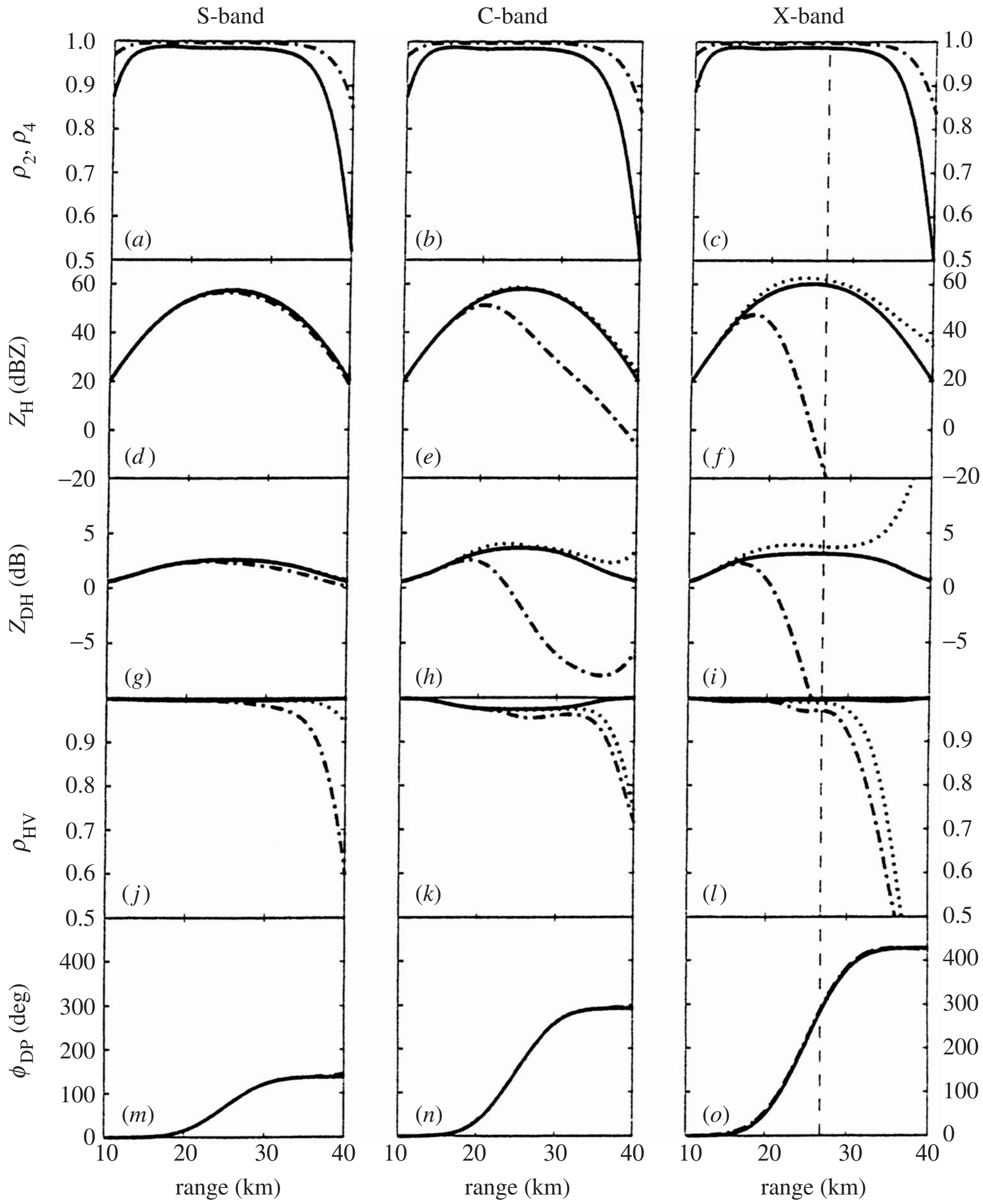

Figure 2. Orientation parameters (ρ), horizontal reflectivity (Z_H) differential reflectivity (Z_{DR}), co-polar correlation coefficient at zero lag time (ρ_{HV}) and two-way total (ϕ) and differential propagation phase shift (ϕ_{DP}) as functions of range, S-, C- and X-band for a model convective rain cell: (a)–(c) solid line, ρ_4; dashed line, ρ_2; (d)–(f) solid line, true intrinsic values; dot-dashed line, measured apparent values; dotted line, estimated values; (m)–(o) solid line, ϕ_{DP}; dashed line, ϕ with ρ_2 as shown in (a)–(c), respectively. The vertical thin dashed line at X-band indicates the range beyond which the radar echoes are lower than the minimum detectable signal (from Torlaschi & Gingras 2000).

(i) the differential reflectivity, $Z_{\mathrm{DR}} = Z_{\mathrm{H}}/Z_{\mathrm{V}}$; Z_{H} = horizontal reflectivity; Z_{V} = vertical reflectivity;

(ii) the specific attenuations of the radar energy as it passes through rain A_{H} (polar H) and A_{V} (polar V) in decibels per kilometre;

(iii) the specific differential phase shift K_{DP} of the electromagnetic waves with different polarizations as they pass through rain in degrees per kilometre (known as the propagation effect); and

(iv) the backscatter phase shift δ of the electromagnetic waves with different polarizations.

Other cross-polar parameters may also be derived from the coherency matrix (Born & Wolf 1964; Ryzhkov 2001), but the above are the parameters which have been examined in the search for methods of measuring rainfall and estimating hydrometeor type.

Testud *et al*. (2000) summarize the rainfall-estimation procedures which have been developed, and which are relatively robust, adding a new method developed for use with the space-borne rain radar of the Tropical Rainfall Measurement Mission (TRMM) but proposed for ground-based polarimetric radars. To these we may add a further technique using Z_{DR} and K_{DP}. Hence, techniques used to retrieve precipitation amount are as follows.

(i) The reflectivity-only method developed over many years assuming the Marshall–Palmer (Marshall & Palmer 1948) drop-size distribution or a gamma distribution (Ulbrich 1983), where

$$R(Z_{\mathrm{H}}) = 2.92 \times 10^{-2} Z_{\mathrm{H}}^{1/1.5}. \tag{3.1}$$

The actual indices vary with precipitation type.

(ii) A two-parameter method as proposed by Chandrasekar *et al*. (1990), where

$$R(Z_{\mathrm{H}}, Z_{\mathrm{DR}}) = 1.98 \times 10^{-3} Z_{\mathrm{H}}^{0.97} Z_{\mathrm{DR}}^{-1.05}. \tag{3.2}$$

(iii) A K_{DP} estimator derived by Sachidananda & Zrnic (1987), where

$$\begin{aligned} R(K_{\mathrm{DP}}) &= 37.1 K_{\mathrm{DP}}^{0.85} \quad \text{(S-band)} \\ &= 16.03 K_{\mathrm{DP}}^{0.95} \quad \text{(C-band)}. \end{aligned} \tag{3.3}$$

(iv) A K_{DP} and Z_{DR} estimator proposed by Ryzhkov & Zrnic (1995):

$$R(K_{\mathrm{DP}}, Z_{\mathrm{DR}}) = 52 K_{\mathrm{DP}}^{0.96} Z_{\mathrm{DR}}^{-0.447}. \tag{3.4}$$

(v) A rain-profiling algorithm (Testud *et al*. 2000) based upon a set of three power-law relationships between A and Z, K_{DP} and A and R and A, each of which is 'normalized' by the N_0^* of the drop-size distribution, where

$$A = a[N_0^*]^{1-b} Z^b, \tag{3.5}$$

$$K_{\mathrm{DP}} = \alpha[N_0^*]^{1-\beta} A^{\beta}, \tag{3.6}$$

$$R = c[N_0^*]^{1-d} A^d, \tag{3.7}$$

where a, α, b, β, c and d are constants dependent upon radar frequency. N_0^* is the intercept parameter of an exponential or gamma drop-size distribution.

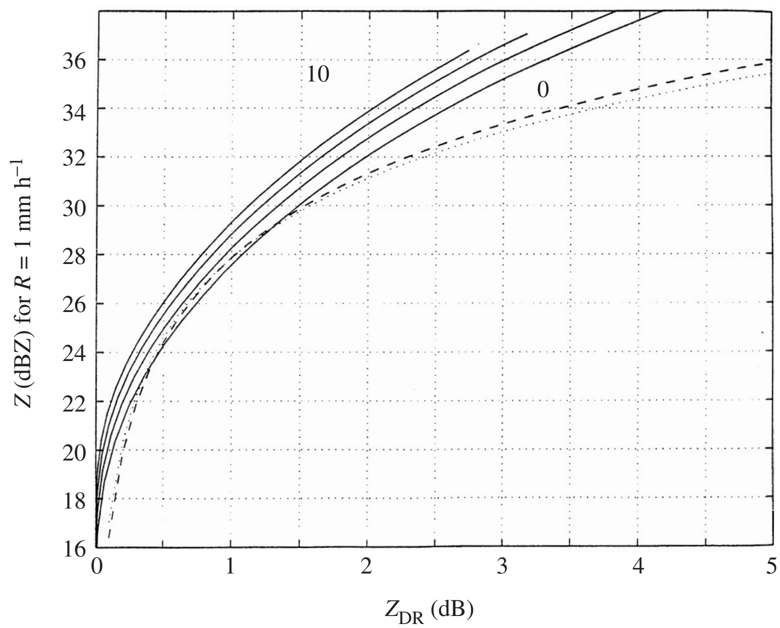

Figure 3. Values of Z which would give a rain rate of 1 mm h^{-1} as a function of Z_{DR} at raindrop spectra with $\mu = 0$, 2, 5 and 10 (solid lines). The actual rain rate scales with the observed Z. Dashed lines from Chandrasekar & Bringi (1988) using 'linear' drop shapes (from Illingworth *et al.* 2000).

Looking first at differential reflectivity, the observed value of this parameter provides an estimate of D_O, the equivolumetric median drop diameter in the gamma drop-size distribution function, which is independent of drop concentration. Figure 3 shows the values of Z which would give a rain rate of $1\ \mathrm{mm\ h^{-1}}$ as a function of Z_{DR} at S-band for raindrop spectra having different values of μ, the index of the gamma function. The curves show that the uncertainty in rainfall rate for a given Z and Z_{DR} is at least ± 1 dB or $\pm 25\%$ and is more likely to be larger than this due to uncertainty in the absolute calibration of Z, contamination by hail, the drop-shape model, the accuracy of Z_{DR} measurements, reflectivity gradients and random effects.

At C-band, attenuation will lead to increasingly negative values of Z_{DR} as the beam traverses heavy rain. Hence, improvements in rainfall estimation are only possible if a very stable and reliable scheme to correct Z_{DR} for attenuation can be implemented. Bringi *et al.* (1990), Smyth & Illingworth (1998) and Carey *et al.* (2000) have proposed procedures.

The use of K_{DP} alone (equation (3.3)) to derive rainfall estimates has several advantages: it is unaffected by attenuation; there is no need for absolute calibration (although a relative calibration between the V and H channels is needed); the relationship to R is almost linear; and it is unaffected by hail (see COST-75 2001; Zrnic *et al.* 2000). However, in practice, there are disadvantages in using K_{DP}, namely, the signal is noisy and small, there is a loss of spatial resolution, sensitivity to ground clutter, problems with backscatter effects and gradients across the beam, ambiguities in stratiform ice and radome effects. Brandes *et al.* (2001) conclude that since bias factors and correlation coefficients between estimated rainfalls and rain-gauge observations for K_{DP} are similar to those of radar reflectivity, there is no obvious benefit in using K_{DP} for a well-calibrated radar. However, K_{DP}-derived rainfall estimates are insensitive to anomalous propagation.

The use of K_{DP} and Z_{DR} ignores the availability of Z values, and if hail is present, Z_{DR} is not the true value for rain. However, Keenan *et al*. (2001) found that $R(K_{\mathrm{DP}}, Z_{\mathrm{DR}})$ are more robust to both drop-size distribution and raindrop axial ratio variations. Goddard *et al*. (1994) showed that Z_{DR} and K_{DP} are not independent and that in rain K_{DP}/Z is a unique function of Z_{DP} that is independent of the μ of the rainfall. They suggested that this redundancy could be exploited to provide an absolute calibration of Z. Smyth *et al*. (1999) took this idea further by suggesting that the best technique in heavy convective rainfall is to continually monitor the consistency of Z and Z_{DR} and the observed value of total phase shift ϕ_{DP} (K_{DP} is the rate of change of ϕ_{DP} with range measured in degrees km^{-1}).

Finally, the rain-profiling algorithm of Testud *et al*. (2000) has been proposed as a method of dealing with complex situations where two, or several, types of rainfall occur along the radar beam, e.g. convective and stratiform rain. The procedure contains a correction scheme for propagation effects at X- (3 cm) or C-band frequencies. However, the algorithm fails when there are large errors in ϕ_{DP} along the integration path.

4. Assimilation of measurements of rainfall into meteorological and hydrological models

(*a*) *Summary of errors in radar rainfall measurements*

As discussed in § 2 *b*, a number of difficulties must be faced in retrieving estimates of surface rainfall from radar. Of particular importance, notably in complex terrain, is the vertical variability of radar reflectivity. However, this source of error can be partly corrected for if the VPR is known.

Vignal *et al*. (2000) discuss three approaches to the determination of the VPR. They find that a correction scheme based on a climatological profile improves the accuracy of daily radar rain estimates significantly within 130 km range of the radar. The fractional standard error (FSE) is reduced from the uncorrected value of 44% to a corrected value of 31%. Further improvement is achieved using a single, mean hourly average VPR (FSE = 25%), and using a locally identified profile (FSE = 23%) (figure 4). This analysis was carried out for both stratiform and convective rainfall, although it was noted that a higher improvement (lower FSE) was obtained for the stratiform events.

Figure 4 also shows the improvement in radar–rain gauge agreement achieved by grouping rain gauges (deviation A in figure), which is similar to that of the profile correction with identified profiles (B). After grouping the rain gauges (i.e. increasing the representativeness of the 'ground truth'), profile corrections are even more effective (improvement C).

It seems clear that a single polarization radar can be used to measure daily rainfall to an accuracy approaching 10%, provided the VPR is adjusted carefully. Such a level of accuracy is similar to that provided by rain gauges. However, sub-daily rainfall is much more problematic, particularly at C-band and shorter wavelengths for which attenuation of the radar beam through heavy rainfall is a serious problem.

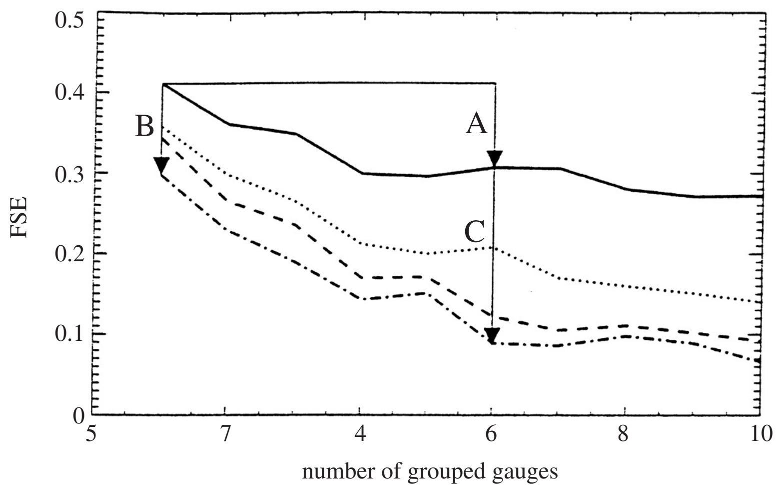

Figure 4. Influence of merging rain-gauge measurements on the comparison between gauges and radar on 14 July 1997 in Switzerland: ——, uncorrected; · · · · ·, climatological VPR; – – –, mean VPR; – · – · –, identified VPR. FSE denotes the fractional standard error (from Vignal *et al*. 2000).

Polarimetric techniques have promised over the last 20 years or so to alleviate, or even remove, some of the difficulties with radar measurements. However, the level of improvement that can be achieved operationally remains unclear. Illingworth (2001) concludes that high-resolution values of Z_{DR} and K_{DP} are probably not accurate enough for use in an operational environment to improve rainfall rates, although the redundancy in these parameters can be used to calibrate Z to 0.5 dB in rain. Negative values of Z_{DR} may enable attenuation at C-band and even X-band to be identified, and the use of Z, Z_{DR} and K_{DP} together may enable hail to be identified reliably. Polarization parameters can identify anomalous propagation echoes.

More work needs to be done to clarify the most reliable method for estimating rainfall using polarization radar. It is clear that the most successful approaches will involve the combination of different parameters. Unfortunately, the variations in the form of the relationships between these parameters may limit the improvements that can be achieved. It is worth reminding ourselves that where growth or evaporation below the lowest radar beam used occurs surface rainfall can only be estimated by recourse to the use of microphysical models and surface observations. Nevertheless, polarimetric radars do offer a way forward to improving rainfall measurements, but hydrologists should not wait for these improvements. They should input these data to hydrological models, albeit taking note that the input has unusual error characteristics. We discuss next how to deal with these error characteristics.

(*b*) *Dealing with measurement errors: use of NWP data in radar analysis*

While work continues on the development of procedures to improve radar estimates of rainfall using the radar data alone, operational systems such as those in the UK are increasingly using the output from mesoscale numerical models (Kitchen *et al*. 1994; Golding 1998). The expectation is that, as the spatial resolution of the models improves, precipitation systems become better resolved and the model analyses and forecasts become more reliable.

It is already clear that some model parameters are predicted more reliably than those deduced from radar data alone. For example, Mittermaier & Illingworth (2002) found that the temperature field in the European Centre for Medium Range Weather Forecasting (ECMWF) and the Met Office models can predict the height of the bright band with a standard deviation of between 150 and 250 m for the entire spectrum of different forecast lead times. They concluded that the model-predicted height of the bright band is probably more accurate than the height derived from radar data themselves.

However, it is unlikely that NWP rainfall output will be of sufficient accuracy to be used in any other than a qualitative way to identify shortcomings in high-resolution observational data. This is in spite of the fact that procedures have been in place for many years to use the output to flag deficiencies in observational systems (see, for example, Lorenc & Hammon 1988).

(*c*) *Dealing with measurement errors: use of complementary data*

Anomalous propagation echoes may be removed where satellite imagery indicates cloud-free skies, and also where nearby stations report the absence of rain as described by Pamment & Conway (1997). The use of propagation link data (§ 2 *b*) shows considerable promise for radar-data quality control.

The use of satellite imagery to estimate rainfall has been reviewed by Rosenfeld & Collier (1998), who concluded that, in most rainfall types other than convective rainfall, satellite techniques are generally rather poor in mid-latitudes. The best accuracy for areal rainfall measurements from space is, at present, obtained over the tropical oceans (see, for example, Atlas & Bell 1992). Ground-based radar offers higher accuracy than satellite techniques over the same time–space domain within the quantitative range of the radar (up to *ca.* 150 km range). Consequently, satellite data are unlikely to be used directly to improve rainfall estimates, whereas rain-gauge data are still likely to be used as described in § 4 *a*.

(*d*) *Assimilation procedures in NWP models*

We have seen that radar measurements of rainfall are likely to have variable error characteristics. Even though new algorithms and radar technology promise improvements, errors will remain just as they exist for other observing systems. However, there is growing impetus to assimilate radar data into NWP models.

NWP is an initial-value problem, and the accuracy of any forecast is directly related to the accuracy of the initial state. Consequently, the assimilation of observations in four dimensions, using the model as a tool, is necessary to determine the initial state. A data-assimilation step consists of a first guess (a 6 h forecast from a previous state, say), an analysis procedure using all available observations within a defined time window, and finally an initialization step.

Radar-data assimilation schemes do not generally make direct use of reflectivity data. Instead they take a precipitation product and use this in the cloud analysis to help in determining the moisture fields. In the UK's operational scheme this is part of the moisture observation preprocessing system (MOPS), which is described in detail by Macpherson *et al.* (1996). Here the radar rainfall estimates provide information on appropriate cloud layers to use within the model, and on whether pixels are wet or dry, thereby influencing the model's relative humidity fields.

Another form of data assimilation is through a latent heat nudging (LHN) scheme, such as that described by Jones & Macpherson (1997). This scheme uses the fact that rainfall rate is proportional to the condensation rate profile within the model. Here temperature increments are introduced slowly through a nudging process. The ratio of observed-to-model precipitation rate is used to scale the profile of latent heat, such that the diagnosed precipitation rate agrees more closely with observations. Unfortunately, the impact of the radar data on the accuracy of the precipitation forecasts weakens considerably beyond *ca.* 12 h ahead.

More comprehensive three- and four-dimensional assimilation schemes have been developed over the last few years (Lorenc *et al*. 2000). Thépaut & Courtier (1991) demonstrated that observing only the small scales of the flow leads to a good reconstruction of both small scales and large scales. This result has been supported by Nuret *et al*. (2000) in a study of the impact of Doppler-radar-derived wind fields on a mesoscale model. Observations of the mass-field evolution also lead to a good reconstruction of the velocity field in mid-latitudes, although less so in the tropics.

It is clear that the improvements in model forecasts are achievable through the assimilation of radar data. However, the benefits may only be realizable if radar data are assimilated together with comprehensive surface data and satellite data, particularly in convective rainfall situations with high-resolution (2.5 km) numerical models (Ducrocq *et al*. 2000). This combination of data ensures that a dynamically consistent initial state is defined, and ensures that errors in the data are held in check. However, the forecasts are found to be very sensitive to the amount of moisture in the initial state, and therefore large errors in the radar data are likely to have a significant impact.

(*e*) *Dealing with measurement errors in hydrological models*

Beven (1996) emphasizes the importance of associating hydrological-model predictions with estimates of predictive uncertainty. He notes two approaches to estimating uncertainty. Firstly, procedures based upon the assumption that the uncertainty derives from that in model parameter values and boundary conditions (see, for example, Storm *et al*. 1998; Bell & Moore 2000). Maximum likelihood or some other parameter-estimation technique is used to find the optimal estimates of model parameters.

Secondly, procedures based upon Monte Carlo simulation using randomly chosen sets of parameter values (for example the generalized likelihood uncertainty estimation (GLUE) methodology outlined by Beven & Binley (1992)). Both these approaches have similarities to the development of ensemble forecasting in meteorology, particularly the latter approach.

In the first approach, one parameter set gives the highest likelihood value, but, as pointed out by Beven (1996), other parameter sets will give values nearly as high, and the distribution of likelihood values will depend on the model calibration period. On the other hand, the second approach uses the distribution of likelihood values as a weighting function to condition the range of predicted variables, i.e. to set uncertainty bounds on, for example, flow predictions. As in meteorology, the likelihood distribution is updated using Bayes equation,

$$L[M(\theta)] = L_0[M/\theta]L_T[M(\theta/Y_T, V_T)]/C,$$

where $L_0([M/\theta)]$ is the prior likelihood for the model $M(\theta)$ associated with parameter vector θ; $L_T[M(\theta/Y_T, V_T)])$ is the likelihood measure calculated for the model over period T with input vector Y_T and observed variable vector V_T, and C is a scaling constant.

The availability of radar measurements of rainfall greatly increased the availability of data for input to hydrological models. Some hydrologists argue that this complicates the procedure for deciding whether a model generates reliable predictions, whereas others argue that this should not be the case if the model is structured to cope with data which are not reliant on calibration processes (Refsgaard *et al*. 1996).

How do radar data fit into this picture? We have seen that the error characteristics can be very variable in space and time, and in spite of a range of adjustment procedures the situation is likely to remain much the same in the future. It seems necessary then to accept this situation, and develop assimilation procedures which lead to the generation of measures of uncertainty associated with flow predictions.

5. Conclusions

The availability of remotely sensed rainfall data, particularly radar data, have provided the input required by distributed flow forecasting models. Unfortunately, the rather variable error characteristics of these data have, until recently, resulted in a reluctance of meteorologists to use them for quantitative rainfall forecasting, and of hydrologists to use them for flood forecasting. Indeed, Beven (1996) suggested that hydrological science is

> awaiting the development of new measurement techniques, especially a method for the rapid assessment of a measure of the heterogeneity of hydrological responses in space, somewhat better than those available from current remote-sensing capability.

The author of this paper is sceptical about such a view as, for the case of rainfall, radar provides the spatial and temporal data required. What is required are comprehensive assimilation procedures which cope with the error characteristics. New measurement techniques may come along. Indeed, the development of polarization-radar techniques in the last few years provides just such a technique. However, no techniques do, or will, provide zero level of error. One might be tempted to suggest that there is a point beyond which it is not productive to keep on trying to remove measurement errors, but it is much more productive to devote resources to coping with errors that exist. Whether we have reached that point yet is unclear. Clearly, there is a balance to be struck. In developing the hydrological aspects of weather prediction and flood warning, Droegemeier *et al*. (2000) conclude that

> because floods are among the most destructive and societally disruptive of all natural phenomena. . . the hydrological and meteorological community [must recognize] their differences, their commonality, and the opportunities resulting from both.

This is particularly true in that part of our science dealing with the use of observational data in forecasting rainfall and the resultant flooding.

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